

```
In [5]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

In [8]: data = pd.read_csv("IMDB-Movie-Data.csv")
```

1. Display Top 10 Rows of The Dataset

```
In [10]: data.head(10)
```

Out[10]:

	Rank	Title	Genre	Description	Director	Actors	Year	Runtime (Minutes)	Rating	Votes	Revenue (Millions)	Metascore
0	1	Guardians of the Galaxy	Action,Adventure,Sci-Fi	A group of intergalactic criminals are forced ...	James Gunn	Chris Pratt, Vin Diesel, Bradley Cooper, Zoe S...	2014	121	8.1	757074	333.13	76.0
1	2	Prometheus	Adventure,Mystery,Sci-Fi	Following clues to the origin of mankind, a te...	Ridley Scott	Noomi Rapace, Logan Marshall-Green, Michael Fa...	2012	124	7.0	485820	126.46	65.0
2	3	Split	Horror,Thriller	Three girls are kidnapped by a man with a diag...	M. Night Shyamalan	James McAvoy, Anya Taylor-Joy, Haley Lu Richar...	2016	117	7.3	157606	138.12	62.0
3	4	Sing	Animation,Comedy,Family	In a city of humanoid animals, a hustling thea...	Christophe Lourdelet	Matthew McConaughey,Reese Witherspoon, Seth Ma...	2016	108	7.2	60545	270.32	59.0
4	5	Suicide Squad	Action,Adventure,Fantasy	A secret government agency recruits some of th...	David Ayer	Will Smith, Jared Leto, Margot Robbie, Viola D...	2016	123	6.2	393727	325.02	40.0
5	6	The Great Wall	Action,Adventure,Fantasy	European mercenaries searching for black powde...	Yimou Zhang	Matt Damon, Tian Jing, Willem Dafoe, Andy Lau	2016	103	6.1	56036	45.13	42.0
6	7	La La Land	Comedy,Drama,Music	A jazz pianist falls for an aspiring actress i...	Damien Chazelle	Ryan Gosling, Emma Stone, Rosemarie DeWitt, J...	2016	128	8.3	258682	151.06	93.0
7	8	Mindhorn	Comedy	A has-been actor best known for playing the ti...	Sean Foley	Essie Davis, Andrea Riseborough, Julian Barrat...	2016	89	6.4	2490	NaN	71.0
8	9	The Lost City of Z	Action,Adventure,Biography	A true-life drama, centering on British explor...	James Gray	Charlie Hunnam, Robert Pattinson, Sienna Mille...	2016	141	7.1	7188	8.01	78.0
9	10	Passengers	Adventure,Drama,Romance	A spacecraft traveling to a distant colony pla...	Morten Tyldum	Jennifer Lawrence, Chris Pratt, Michael Sheen,...	2016	116	7.0	192177	100.01	41.0

2. Check Last 10 Rows of The Dataset

```
In [11]: data.tail(10)
```

Out[11]:

	Rank	Title	Genre	Description	Director	Actors	Year	Runtime (Minutes)	Rating	Votes	Revenue (Millions)	Metascore
990	991	Underworld: Rise of the Lycans	Action,Adventure,Fantasy	An origins story centered on the centuries-old...	Patrick Tatopoulos	Rhona Mitra, Michael Sheen, Bill Nighy, Steven...	2009	92	6.6	129708	45.80	44.0
991	992	Taare Zameen Par	Drama,Family,Music	An eight-year-old boy is thought to be a lazy ...	Aamir Khan	Darsheel Safary, Aamir Khan, Tanay Chheda, Sac...	2007	165	8.5	102697	1.20	42.0
992	993	Take Me Home Tonight	Comedy,Drama,Romance	Four years after graduation, an awkward high s...	Michael Dowse	Topher Grace, Anna Faris, Dan Fogler, Teresa P...	2011	97	6.3	45419	6.92	NaN
993	994	Resident Evil: Afterlife	Action,Adventure,Horror	While still out to destroy the evil Umbrella C...	Paul W.S. Anderson	Milla Jovovich, Ali Larter, Wentworth Miller,K...	2010	97	5.9	140900	60.13	37.0
994	995	Project X	Comedy	3 high school seniors throw a birthday party t...	Nima Nourizadeh	Thomas Mann, Oliver Cooper, Jonathan Daniel Br...	2012	88	6.7	164088	54.72	48.0
995	996	Secret in Their Eyes	Crime,Drama,Mystery	A tight-knit team of rising investigators, alo...	Billy Ray	Chiwetel Ejiofor, Nicole Kidman, Julia Roberts...	2015	111	6.2	27585	NaN	45.0
996	997	Hostel: Part II	Horror	Three American college students studying abroa...	Eli Roth	Lauren German, Heather Matarazzo, Bijou Philli...	2007	94	5.5	73152	17.54	46.0
997	998	Step Up 2: The Streets	Drama,Music,Romance	Romantic sparks occur between two dance studen...	Jon M. Chu	Robert Hoffman, Briana Evigan, Cassie Ventura,...	2008	98	6.2	70699	58.01	50.0
998	999	Search Party	Adventure,Comedy	A pair of friends embark on a mission to reuni...	Scot Armstrong	Adam Pally, T.J. Miller, Thomas Middleditch,Sh...	2014	93	5.6	4881	NaN	22.0
999	1000	Nine Lives	Comedy,Family,Fantasy	A stuffy businessman finds himself trapped ins...	Barry Sonnenfeld	Kevin Spacey, Jennifer Garner, Robbie Amell,Ch...	2016	87	5.3	12435	19.64	11.0

3. Find Shape of Our Dataset (Number of Rows And Number of Columns)

In [12]: data.shape

Out[12]: (1000, 12)

In [13]: print('Number of Rows',data.shape[0])
print('Number of Columns',data.shape[1])

Number of Rows 1000
Number of Columns 12

4. Getting Information About Our Dataset Like Total Number Rows, Total Number of Columns, Datatypes of Each Column And Memory Requirement

In [14]: data.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 12 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Rank                1000 non-null   int64
1   Title               1000 non-null   object
2   Genre               1000 non-null   object
3   Description          1000 non-null   object
4   Director            1000 non-null   object
5   Actors              1000 non-null   object
6   Year                1000 non-null   int64
7   Runtime (Minutes)   1000 non-null   int64
8   Rating              1000 non-null   float64
9   Votes               1000 non-null   int64
10  Revenue (Millions)  872 non-null    float64
11  Metascore           936 non-null    float64
dtypes: float64(3), int64(4), object(5)
memory usage: 93.9+ KB
```

5. Check Null Values In The Dataset

```
In [16]: print("any missing value?", data.isnull().values.any())
```

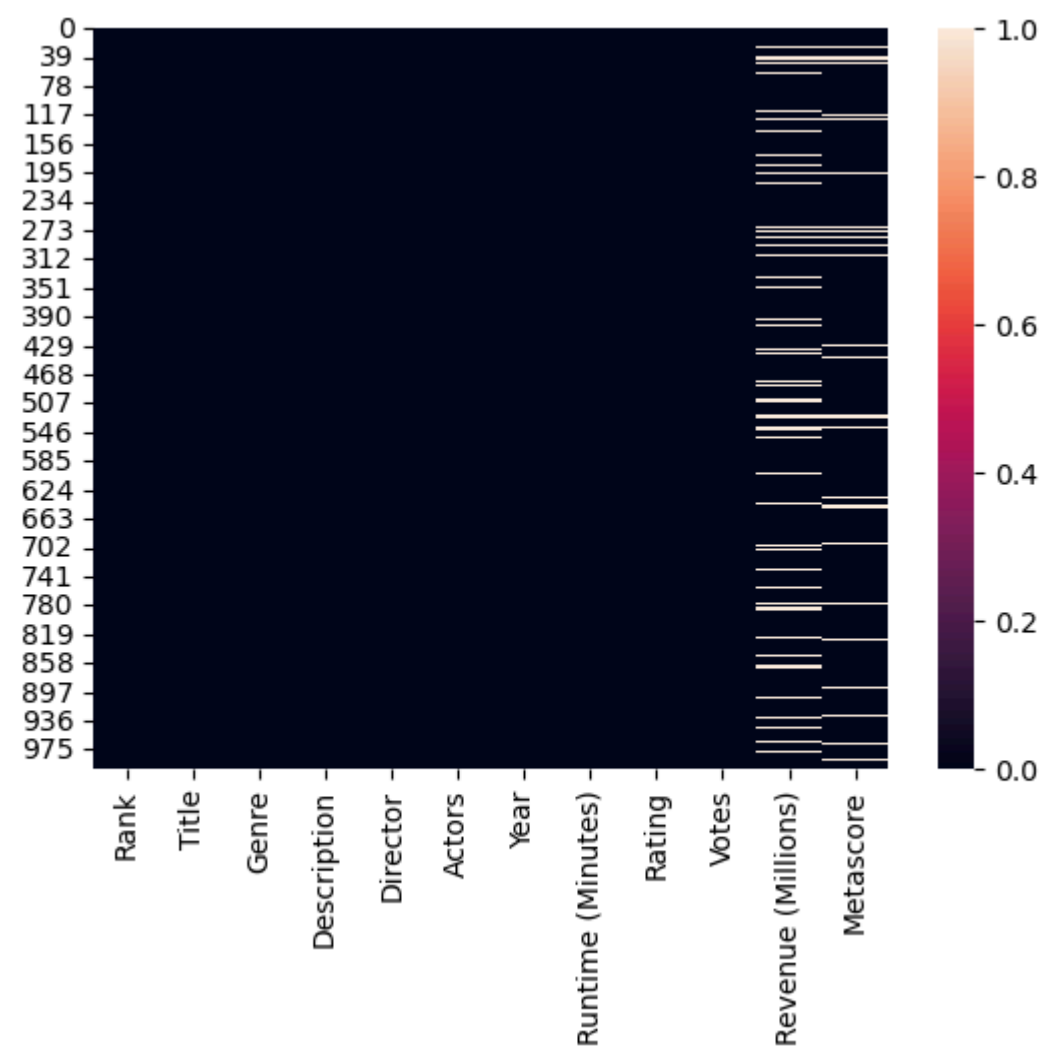
any missing value? True

```
In [18]: data.isnull().sum()
```

```
Out[18]: Rank                0
Title                0
Genre                0
Description          0
Director            0
Actors              0
Year                0
Runtime (Minutes)   0
Rating              0
Votes               0
Revenue (Millions)  128
Metascore           64
dtype: int64
```

```
In [19]: sns.heatmap(data.isnull())
```

Out[19]: <Axes: >



```
In [21]: per_missing = data.isnull().sum() * 100 / len(data)
per_missing
```

```
Out[21]: Rank          0.0
Title            0.0
Genre            0.0
Description       0.0
Director         0.0
Actors           0.0
Year             0.0
Runtime (Minutes) 0.0
Rating           0.0
Votes            0.0
Revenue (Millions) 12.8
Metascore        6.4
dtype: float64
```

6. Drop All The Missing Values

```
In [23]: data.dropna(axis=0)
```

Out[23]:

	Rank	Title	Genre	Description	Director	Actors	Year	Runtime (Minutes)	Rating	Votes	Revenue (Millions)	Metascore
0	1	Guardians of the Galaxy	Action,Adventure,Sci-Fi	A group of intergalactic criminals are forced ...	James Gunn	Chris Pratt, Vin Diesel, Bradley Cooper, Zoe S...	2014	121	8.1	757074	333.13	76.0
1	2	Prometheus	Adventure,Mystery,Sci-Fi	Following clues to the origin of mankind, a te...	Ridley Scott	Noomi Rapace, Logan Marshall-Green, Michael Fa...	2012	124	7.0	485820	126.46	65.0
2	3	Split	Horror,Thriller	Three girls are kidnapped by a man with a diag...	M. Night Shyamalan	James McAvoy, Anya Taylor-Joy, Haley Lu Richar...	2016	117	7.3	157606	138.12	62.0
3	4	Sing	Animation,Comedy,Family	In a city of humanoid animals, a hustling thea...	Christophe Lourdelet	Matthew McConaughey,Reese Witherspoon, Seth Ma...	2016	108	7.2	60545	270.32	59.0
4	5	Suicide Squad	Action,Adventure,Fantasy	A secret government agency recruits some of th...	David Ayer	Will Smith, Jared Leto, Margot Robbie, Viola D...	2016	123	6.2	393727	325.02	40.0
...	...	...	...	...	...	...	...	...	...	...	...	...
993	994	Resident Evil: Afterlife	Action,Adventure,Horror	While still out to destroy the evil Umbrella C...	Paul W.S. Anderson	Milla Jovovich, Ali Larter, Wentworth Miller,K...	2010	97	5.9	140900	60.13	37.0
994	995	Project X	Comedy	3 high school seniors throw a birthday party t...	Nima Nourizadeh	Thomas Mann, Oliver Cooper, Jonathan Daniel Br...	2012	88	6.7	164088	54.72	48.0
996	997	Hostel: Part II	Horror	Three American college students studying abroa...	Eli Roth	Lauren German, Heather Matarazzo, Bijou Philli...	2007	94	5.5	73152	17.54	46.0
997	998	Step Up 2: The Streets	Drama,Music,Romance	Romantic sparks occur between two dance studen...	Jon M. Chu	Robert Hoffman, Briana Evigan, Cassie Ventura,...	2008	98	6.2	70699	58.01	50.0
999	1000	Nine Lives	Comedy,Family,Fantasy	A stuffy businessman finds himself trapped ins...	Barry Sonnenfeld	Kevin Spacey, Jennifer Garner, Robbie Amell,Ch...	2016	87	5.3	12435	19.64	11.0

838 rows × 12 columns

7. Check For Duplicate Data

```
In [24]: dup_data=data.duplicated().any()
print("Are there any duplicated values in data?",dup_data)
```

Are there any duplicated values in data? False

```
In [25]: data=data.drop_duplicates()
data
```

Out[25]:

	Rank	Title	Genre	Description	Director	Actors	Year	Runtime (Minutes)	Rating	Votes	Revenue (Millions)	Metascore
0	1	Guardians of the Galaxy	Action,Adventure,Sci-Fi	A group of intergalactic criminals are forced ...	James Gunn	Chris Pratt, Vin Diesel, Bradley Cooper, Zoe S...	2014	121	8.1	757074	333.13	76.0
1	2	Prometheus	Adventure,Mystery,Sci-Fi	Following clues to the origin of mankind, a te...	Ridley Scott	Noomi Rapace, Logan Marshall-Green, Michael Fa...	2012	124	7.0	485820	126.46	65.0
2	3	Split	Horror,Thriller	Three girls are kidnapped by a man with a diag...	M. Night Shyamalan	James McAvoy, Anya Taylor-Joy, Haley Lu Richar...	2016	117	7.3	157606	138.12	62.0
3	4	Sing	Animation,Comedy,Family	In a city of humanoid animals, a hustling thea...	Christophe Lourdelet	Matthew McConaughey,Reese Witherspoon, Seth Ma...	2016	108	7.2	60545	270.32	59.0
4	5	Suicide Squad	Action,Adventure,Fantasy	A secret government agency recruits some of th...	David Ayer	Will Smith, Jared Leto, Margot Robbie, Viola D...	2016	123	6.2	393727	325.02	40.0
...	...	...	...	...	...	...	...	...	...	...	...	...
993	994	Resident Evil: Afterlife	Action,Adventure,Horror	While still out to destroy the evil Umbrella C...	Paul W.S. Anderson	Milla Jovovich, Ali Larter, Wentworth Miller,K...	2010	97	5.9	140900	60.13	37.0
994	995	Project X	Comedy	3 high school seniors throw a birthday party t...	Nima Nourizadeh	Thomas Mann, Oliver Cooper, Jonathan Daniel Br...	2012	88	6.7	164088	54.72	48.0
996	997	Hostel: Part II	Horror	Three American college students studying abroa...	Eli Roth	Lauren German, Heather Matarazzo, Bijou Philli...	2007	94	5.5	73152	17.54	46.0
997	998	Step Up 2: The Streets	Drama,Music,Romance	Romantic sparks occur between two dance studen...	Jon M. Chu	Robert Hoffman, Briana Evigan, Cassie Ventura,...	2008	98	6.2	70699	58.01	50.0
999	1000	Nine Lives	Comedy,Family,Fantasy	A stuffy businessman finds himself trapped ins...	Barry Sonnenfeld	Kevin Spacey, Jennifer Garner, Robbie Amell,Ch...	2016	87	5.3	12435	19.64	11.0

838 rows × 12 columns

8. Get Overall Statistics About The DataFrame

In [27]: data.describe(include='all')

Out[27]:

	Rank	Title	Genre	Description	Director	Actors	Year	Runtime (Minutes)	Rating	Votes	Revenue (Millions)	Metascore
count	838.000000	838	838	838	838	838	838.00000	838.000000	838.000000	8.380000e+02	838.000000	838.000000
unique	NaN	837	189	838	524	834	NaN	NaN	NaN	NaN	NaN	NaN
top	NaN	The Host	Action,Adventure,Sci-Fi	A group of intergalactic criminals are forced ...	Ridley Scott	Jennifer Lawrence, Josh Hutcherson, Liam Hemsw...	NaN	NaN	NaN	NaN	NaN	NaN
freq	NaN	2	50	1	8	2	NaN	NaN	NaN	NaN	NaN	NaN
mean	485.247017	NaN	NaN	NaN	NaN	NaN	2012.50716	114.638425	6.814320	1.932303e+05	84.564558	59.575179
std	286.572065	NaN	NaN	NaN	NaN	NaN	3.17236	18.470922	0.877754	1.930990e+05	104.520227	16.952416
min	1.000000	NaN	NaN	NaN	NaN	NaN	2006.00000	66.000000	1.900000	1.780000e+02	0.000000	11.000000
25%	238.250000	NaN	NaN	NaN	NaN	NaN	2010.00000	101.000000	6.300000	6.127650e+04	13.967500	47.000000
50%	475.500000	NaN	NaN	NaN	NaN	NaN	2013.00000	112.000000	6.900000	1.368795e+05	48.150000	60.000000
75%	729.750000	NaN	NaN	NaN	NaN	NaN	2015.00000	124.000000	7.500000	2.710830e+05	116.800000	72.000000
max	1000.000000	NaN	NaN	NaN	NaN	NaN	2016.00000	187.000000	9.000000	1.791916e+06	936.630000	100.000000

9. Display Title of The Movie Having Runtime >= 180 Minutes

In [29]: data.columns

Out[29]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year', 'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)', 'Metascore'], dtype='object')

In [28]: data[data['Runtime (Minutes)']>=180]['Title']

Out[28]: 82 The Wolf of Wall Street
88 The Hateful Eight
311 La vie d'Adèle
Name: Title, dtype: object

10. In Which Year There Was The Highest Voting?

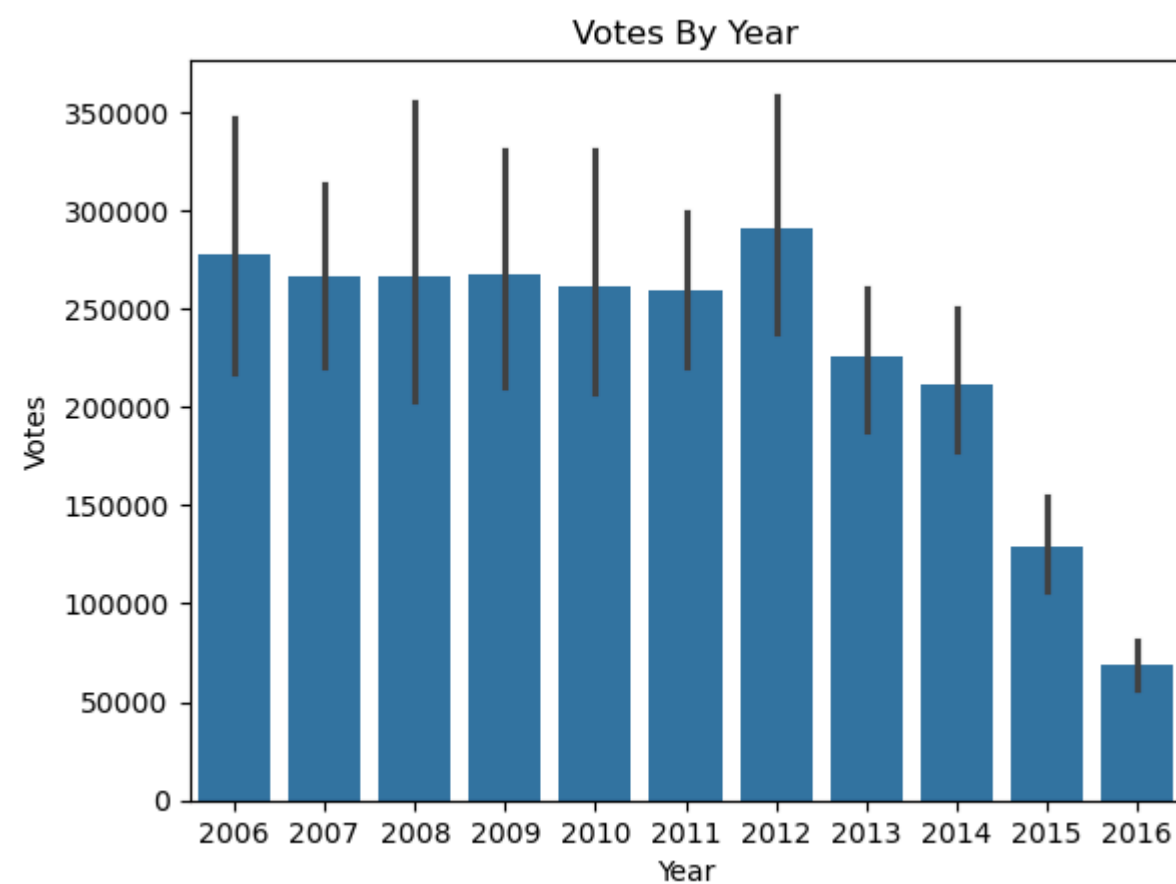
In [30]: data.columns

Out[30]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year', 'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)', 'Metascore'], dtype='object')

In [34]: data.groupby('Year')['Votes'].mean().sort_values(ascending=False)

```
Out[34]: Year
2012    290861.483871
2006    277232.219512
2009    267180.577778
2008    266580.145833
2007    266530.704545
2010    261082.929825
2011    259254.736842
2013    225531.892857
2014    211926.881720
2015    129512.651376
2016     68437.823232
Name: Votes, dtype: float64
```

```
In [35]: sns.barplot(x='Year',y='Votes',data=data)
plt.title("Votes By Year")
plt.show()
```



11. In Which Year There Was The Highest Revenue?

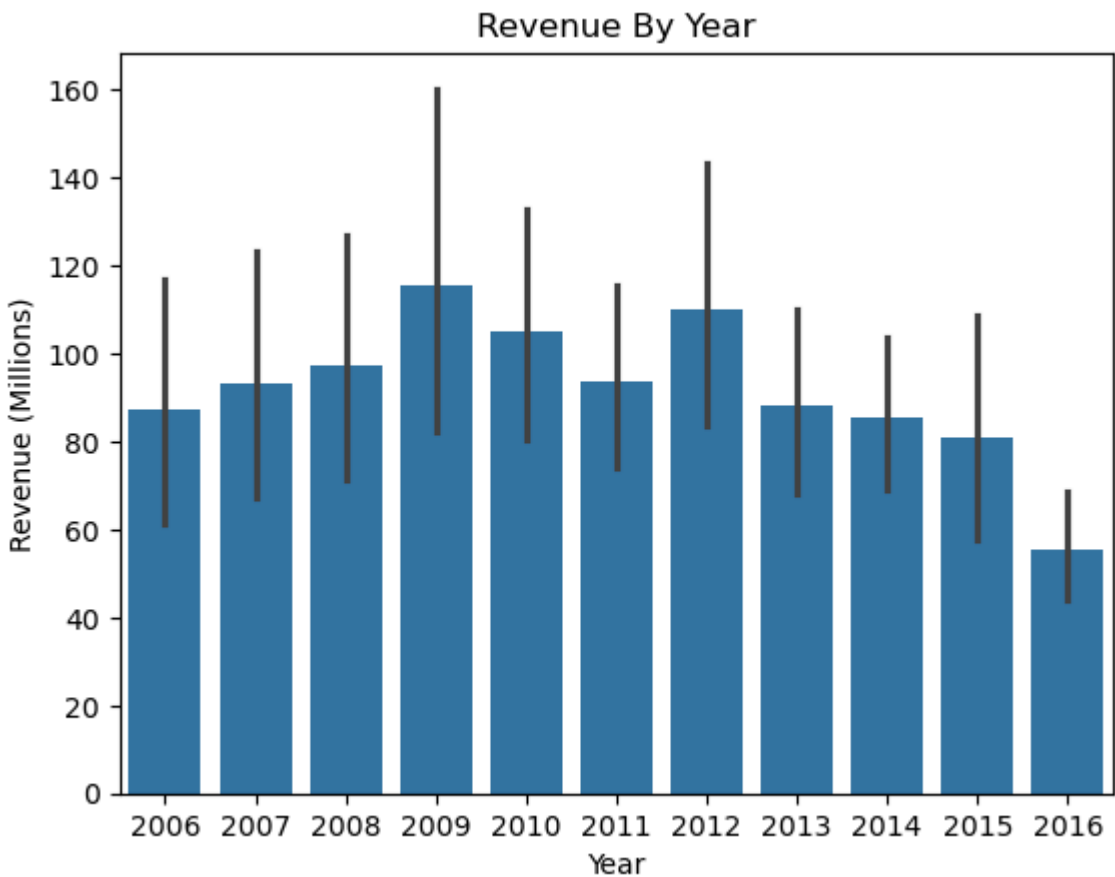
```
In [36]: data.columns
```

```
Out[36]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',
              'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',
              'Metascore'],
              dtype='object')
```

```
In [37]: data.groupby('Year')['Revenue (Millions)'].mean().sort_values(ascending=False)
```

```
Out[37]: Year
2009      115.742000
2012      110.103065
2010      105.081579
2008       97.525417
2011       93.703333
2007       93.074091
2013       88.084643
2006       87.255610
2014       85.433656
2015       80.725596
2016       55.566111
Name: Revenue (Millions), dtype: float64
```

```
In [38]: sns.barplot(x='Year',y='Revenue (Millions)',data=data)
plt.title("Revenue By Year")
plt.show()
```



12. Find The Average Rating For Each Director

```
In [39]: data.columns
```

```
Out[39]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',
              'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',
              'Metascore'],
              dtype='object')
```

```
In [40]: data.groupby('Director')['Rating'].mean().sort_values(ascending=False)
```

```
Out[40]: Director
Christopher Nolan      8.68
Olivier Nakache        8.60
Makoto Shinkai         8.60
Florian Henckel von Donnersmarck  8.50
Aamir Khan             8.50
...
Sam Taylor-Johnson     4.10
Joey Curtis            4.00
George Nolfi           3.90
James Wong             2.70
Jason Friedberg        1.90
Name: Rating, Length: 524, dtype: float64
```

13. Display Top 10 Lengthy Movies Title

```
In [41]: data.columns
```

```
Out[41]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',
              'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',
              'Metascore'],
              dtype='object')
```

```
In [54]: top10_len=data.nlargest(10,'Runtime (Minutes)')[['Title','Runtime (Minutes)']\
        .set_index('Title')
```

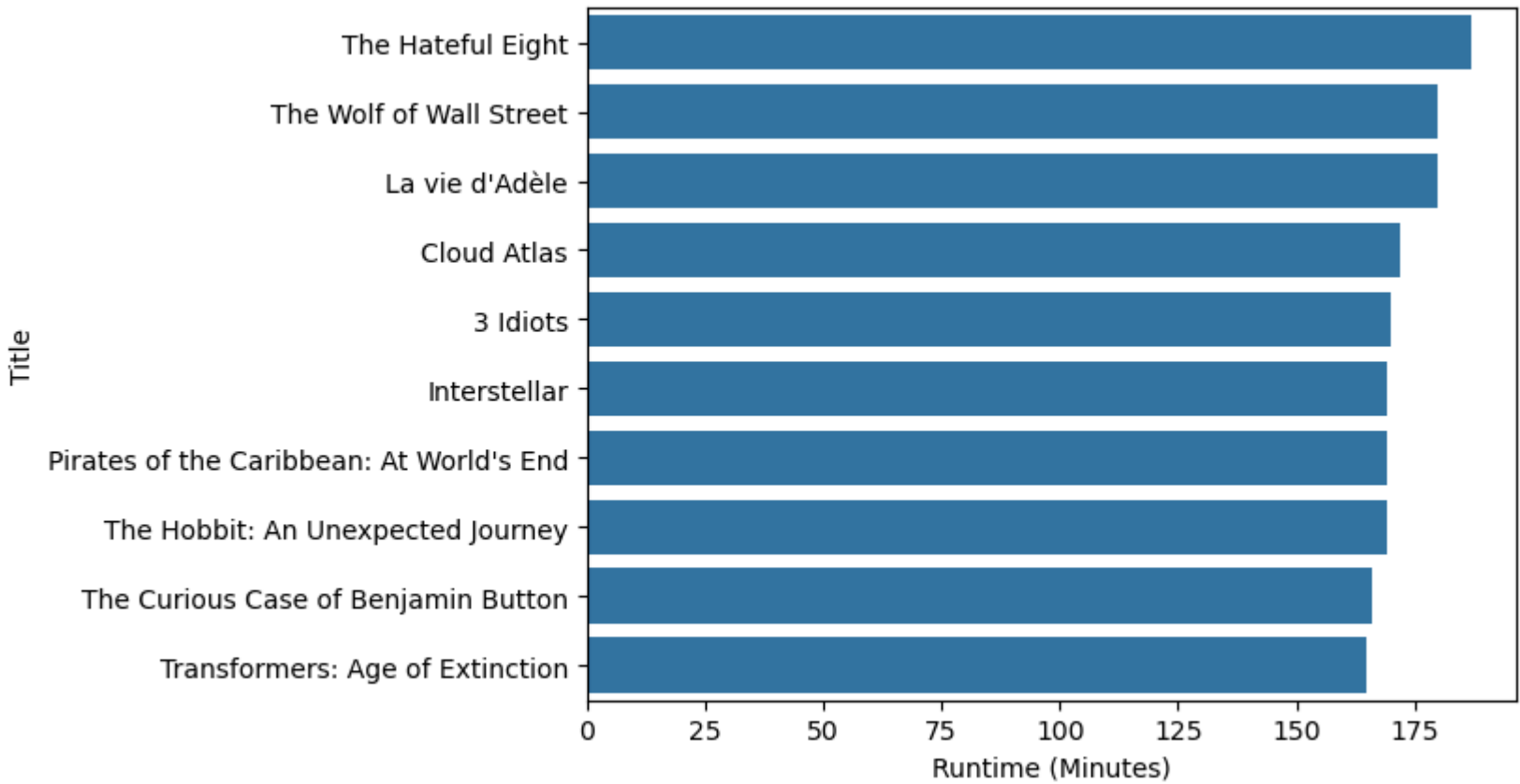
```
In [55]: top10_len
```

Out[55]:

Runtime (Minutes)	
Title	
The Hateful Eight	187
The Wolf of Wall Street	180
La vie d'Adèle	180
Cloud Atlas	172
3 Idiots	170
Interstellar	169
Pirates of the Caribbean: At World's End	169
The Hobbit: An Unexpected Journey	169
The Curious Case of Benjamin Button	166
Transformers: Age of Extinction	165

```
In [57]: sns.barplot(x='Runtime (Minutes)',y=top10_len.index,data=top10_len)
```

```
Out[57]: <Axes: xlabel='Runtime (Minutes)', ylabel='Title'>
```



14. Display Number of Movies Per Year

```
In [58]: data.columns
```

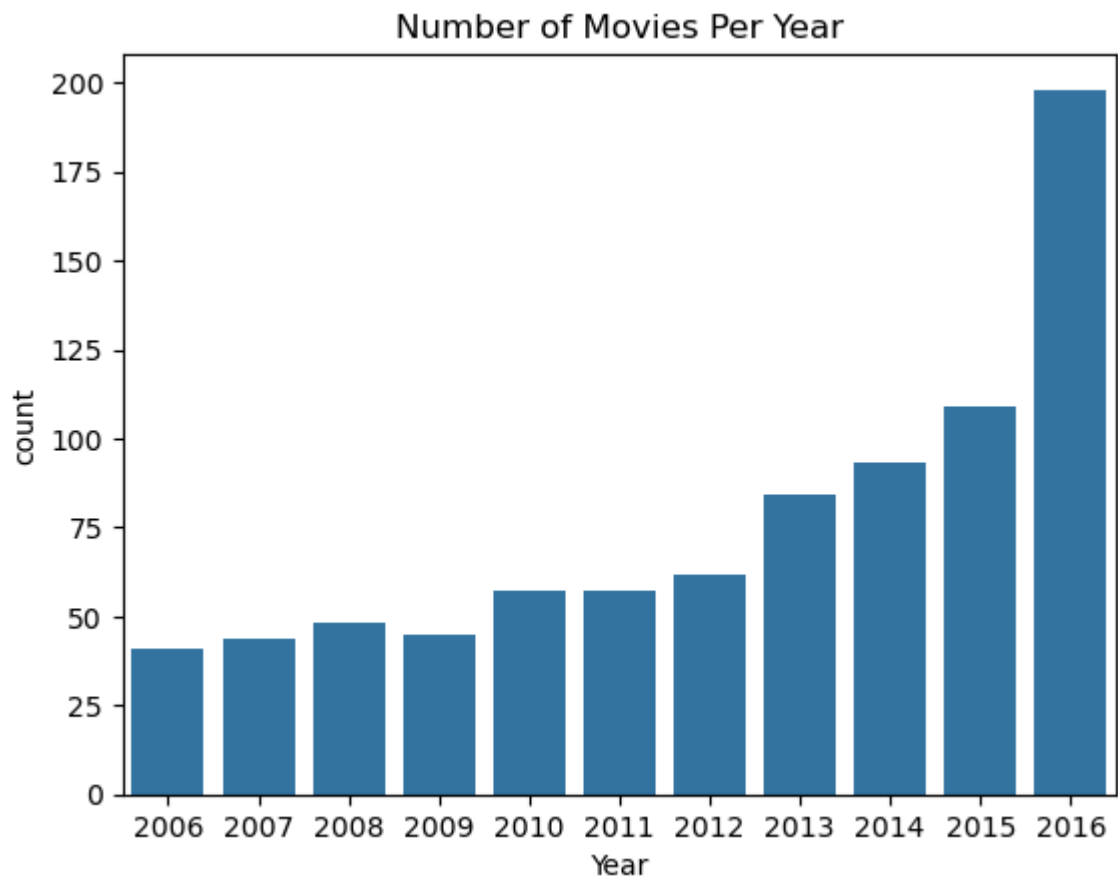
```
Out[58]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',  
              'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',  
              'Metascore'],  
              dtype='object')
```

```
In [60]: data['Year'].value_counts()
```

```
Out[60]: Year  
2016      198  
2015      109  
2014       93  
2013       84  
2012       62  
2011       57  
2010       57  
2008       48  
2009       45  
2007       44  
2006       41  
Name: count, dtype: int64
```

```
In [61]: sns.countplot(x='Year',data=data)  
plt.title("Number of Movies Per Year")
```

```
Out[61]: Text(0.5, 1.0, 'Number of Movies Per Year')
```



```
In [ ]:
```

15. Find Most Popular Movie Title (Highest Revenue)

```
In [62]: data.columns
```

```
Out[62]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',
              'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',
              'Metascore'],
              dtype='object')
```

```
In [63]: data[data['Revenue (Millions)'].max() == data['Revenue (Millions)']]['Title']
```

```
Out[63]: 50    Star Wars: Episode VII - The Force Awakens
Name: Title, dtype: object
```

16. Display Top 10 Highest Rated Movie Titles And its Directors

```
In [64]: data.columns
```

```
Out[64]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',
              'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',
              'Metascore'],
              dtype='object')
```

```
In [65]: top_10=data.nlargest(10,'Rating')[['Title','Rating','Director']].set_index('Title')
```

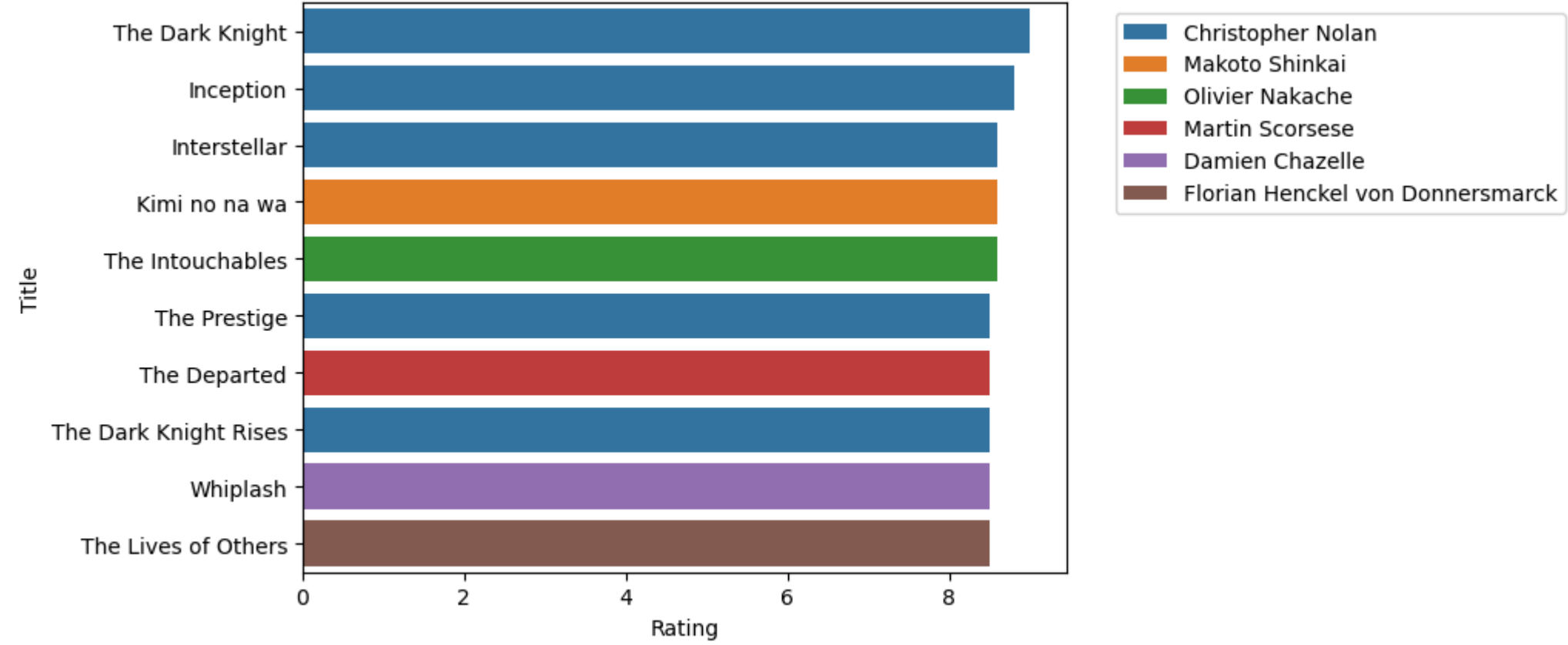
In [66]: top_10

Out[66]:

	Rating	Director
Title		
The Dark Knight	9.0	Christopher Nolan
Inception	8.8	Christopher Nolan
Interstellar	8.6	Christopher Nolan
Kimi no na wa	8.6	Makoto Shinkai
The Intouchables	8.6	Olivier Nakache
The Prestige	8.5	Christopher Nolan
The Departed	8.5	Martin Scorsese
The Dark Knight Rises	8.5	Christopher Nolan
Whiplash	8.5	Damien Chazelle
The Lives of Others	8.5	Florian Henckel von Donnersmarck

In [75]: sns.barplot(x='Rating',y=top10_len.index,data=top10_len,hue='Director',dodge=False)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

Out[75]: <matplotlib.legend.Legend at 0x1262f660f50>



17. Display Top 10 Highest Revenue Movie Titles

```
In [76]: data.columns
```

```
Out[76]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',  
              'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',  
              'Metascore'],  
              dtype='object')
```

```
In [78]: data.nlargest(10,'Revenue (Millions)') ['Title']
```

```
Out[78]: 50      Star Wars: Episode VII - The Force Awakens  
87              Avatar  
85              Jurassic World  
76              The Avengers  
54              The Dark Knight  
12              Rogue One  
119             Finding Dory  
94      Avengers: Age of Ultron  
124             The Dark Knight Rises  
578             The Hunger Games: Catching Fire  
Name: Title, dtype: object
```

```
In [82]: data.sort_values(by='Revenue (Millions)',ascending=False).head(10)
```

Rank		Title	Genre	Description	Director	Actors	Year	Runtime (Minutes)	Rating	Votes	Revenue (Millions)	Metascore
50	51	Star Wars: Episode VII - The Force Awakens	Action,Adventure,Fantasy	Three decades after the defeat of the Galactic...	J.J. Abrams	Daisy Ridley, John Boyega, Oscar Isaac, Domhna...	2015	136	8.1	661608	936.63	81.0
87	88	Avatar	Action,Adventure,Fantasy	A paraplegic marine dispatched to the moon Pan...	James Cameron	Sam Worthington, Zoe Saldana, Sigourney Weaver...	2009	162	7.8	935408	760.51	83.0
85	86	Jurassic World	Action,Adventure,Sci-Fi	A new theme park, built on the original site o...	Colin Trevorrow	Chris Pratt, Bryce Dallas Howard, Ty Simpkins,...	2015	124	7.0	455169	652.18	59.0
76	77	The Avengers	Action,Sci-Fi	Earth's mightiest heroes must come together an...	Joss Whedon	Robert Downey Jr., Chris Evans, Scarlett Johan...	2012	143	8.1	1045588	623.28	69.0
54	55	The Dark Knight	Action,Crime,Drama	When the menace known as the Joker wreaks havo...	Christopher Nolan	Christian Bale, Heath Ledger, Aaron Eckhart,Mi...	2008	152	9.0	1791916	533.32	82.0
12	13	Rogue One	Action,Adventure,Sci-Fi	The Rebel Alliance makes a risky move to steal...	Gareth Edwards	Felicity Jones, Diego Luna, Alan Tudyk, Donnie...	2016	133	7.9	323118	532.17	65.0
119	120	Finding Dory	Animation,Adventure,Comedy	The friendly but forgetful blue tang fish, Dor...	Andrew Stanton	Ellen DeGeneres, Albert Brooks,Ed O'Neill, Kai...	2016	97	7.4	157026	486.29	77.0
94	95	Avengers: Age of Ultron	Action,Adventure,Sci-Fi	When Tony Stark and Bruce Banner try to jump-s...	Joss Whedon	Robert Downey Jr., Chris Evans, Mark Ruffalo, ...	2015	141	7.4	516895	458.99	66.0
124	125	The Dark Knight Rises	Action,Thriller	Eight years after the Joker's reign of anarchy...	Christopher Nolan	Christian Bale, Tom Hardy, Anne Hathaway,Gary ...	2012	164	8.5	1222645	448.13	78.0
578	579	The Hunger Games: Catching Fire	Action,Adventure,Mystery	Katniss Everdeen and Peeta Mellark become targ...	Francis Lawrence	Jennifer Lawrence, Josh Hutcherson, Liam Hemswo...	2013	146	7.6	525646	424.65	76.0

```
In [83]: top_10=data.nlargest(10,'Revenue (Millions)')[['Title','Revenue (Millions)']].set_index('Title')
```

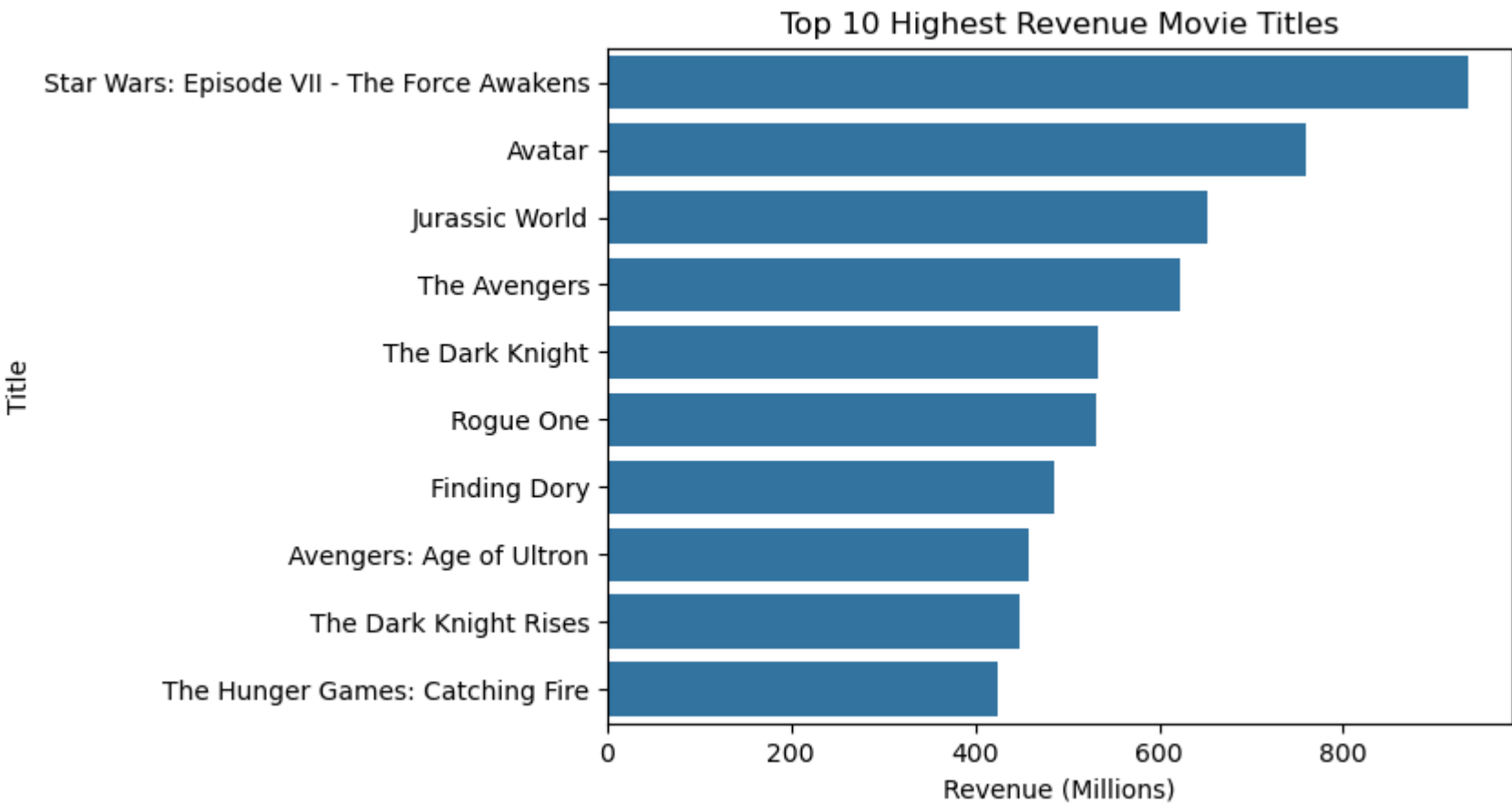
```
In [84]: top_10
```

Out[84]:

	Revenue (Millions)
Star Wars: Episode VII - The Force Awakens	936.63
Avatar	760.51
Jurassic World	652.18
The Avengers	623.28
The Dark Knight	533.32
Rogue One	532.17
Finding Dory	486.29
Avengers: Age of Ultron	458.99
The Dark Knight Rises	448.13
The Hunger Games: Catching Fire	424.65

Title	
Star Wars: Episode VII - The Force Awakens	936.63
Avatar	760.51
Jurassic World	652.18
The Avengers	623.28
The Dark Knight	533.32
Rogue One	532.17
Finding Dory	486.29
Avengers: Age of Ultron	458.99
The Dark Knight Rises	448.13
The Hunger Games: Catching Fire	424.65

```
In [91]: sns.barplot(x='Revenue (Millions)',y=top_10.index,data=top_10)
plt.title("Top 10 Highest Revenue Movie Titles")
plt.show()
```



18. Find Average Rating of Movies Year-wise

```
In [90]: data.columns
```

```
Out[90]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',
              'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',
              'Metascore'],
              dtype='object')
```

```
In [95]: data.groupby('Year')['Rating'].mean().sort_values(ascending=False)
```

```
Out[95]: Year
2006      7.143902
2007      7.140909
2011      6.945614
2012      6.933871
2009      6.911111
2010      6.894737
2013      6.832143
2014      6.822581
2008      6.708333
2015      6.674312
2016      6.644444
Name: Rating, dtype: float64
```

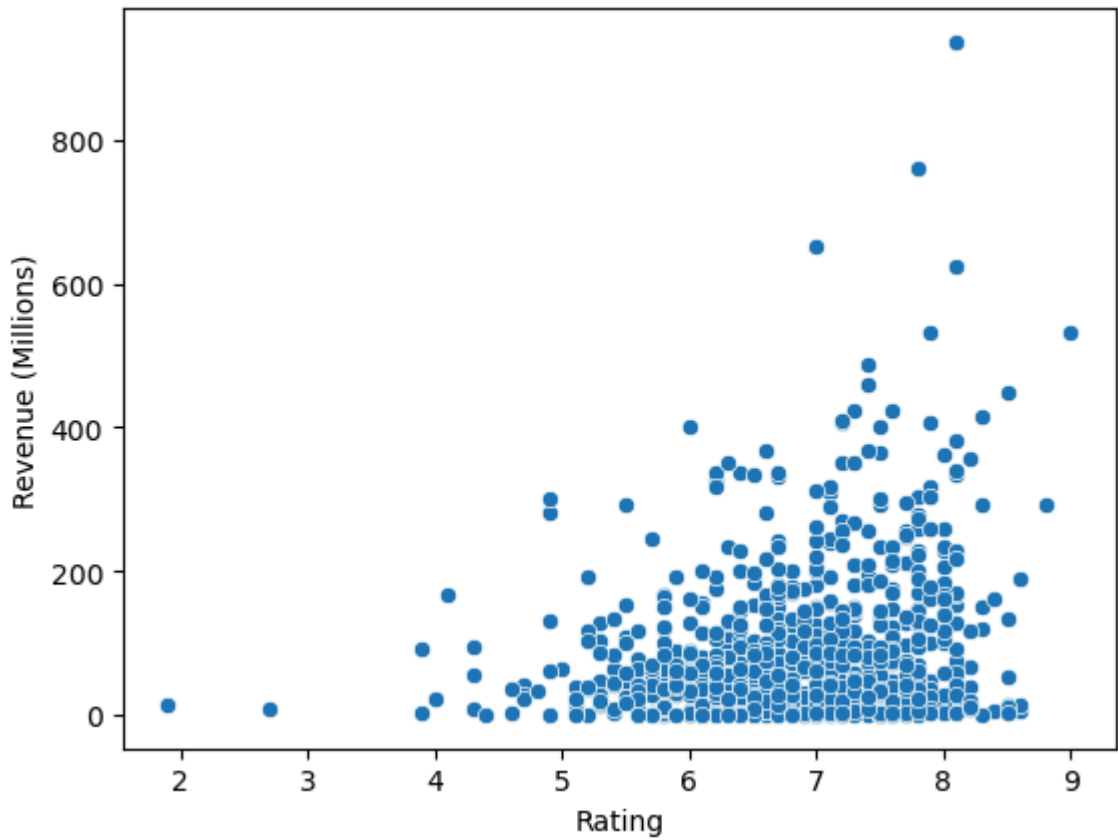
19. Does Rating Affect The Revenue?

```
In [96]: data.columns
```

```
Out[96]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',
              'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',
              'Metascore'],
              dtype='object')
```

```
In [97]: sns.scatterplot(x='Rating',y='Revenue (Millions)',data=data)
```

```
Out[97]: <Axes: xlabel='Rating', ylabel='Revenue (Millions)'>
```



20. Classify Movies Based on Ratings [Good,Better and Best]

```
In [98]: data.columns
```

```
Out[98]: Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',  
              'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',  
              'Metascore'],  
            dtype='object')
```

```
In [99]: def rating(rating):  
         if rating>=7.0:  
             return 'Excellent'  
         elif rating>=6.0:  
             return 'Good'  
         else:  
             return 'Average'
```

```
In [100]: data['rating_cat']=data['Rating'].apply(rating)
```

```
In [102]: data.head()
```

Out[102...

	Rank	Title	Genre	Description	Director	Actors	Year	Runtime (Minutes)	Rating	Votes	Revenue (Millions)	Metascore	rating_cat
0	1	Guardians of the Galaxy	Action,Adventure,Sci-Fi	A group of intergalactic criminals are forced ...	James Gunn	Chris Pratt, Vin Diesel, Bradley Cooper, Zoe S...	2014	121	8.1	757074	333.13	76.0	Excellent
1	2	Prometheus	Adventure,Mystery,Sci-Fi	Following clues to the origin of mankind, a te...	Ridley Scott	Noomi Rapace, Logan Marshall-Green, Michael Fa...	2012	124	7.0	485820	126.46	65.0	Excellent
2	3	Split	Horror,Thriller	Three girls are kidnapped by a man with a diag...	M. Night Shyamalan	James McAvoy, Anya Taylor-Joy, Haley Lu Richar...	2016	117	7.3	157606	138.12	62.0	Excellent
3	4	Sing	Animation,Comedy,Family	In a city of humanoid animals, a hustling thea...	Christophe Lourdelet	Matthew McConaughey,Reese Witherspoon, Seth Ma...	2016	108	7.2	60545	270.32	59.0	Excellent
4	5	Suicide Squad	Action,Adventure,Fantasy	A secret government agency recruits some of th...	David Ayer	Will Smith, Jared Leto, Margot Robbie, Viola D...	2016	123	6.2	393727	325.02	40.0	Good

21. Count Number of Action Movies

In [116...

```
data.columns
```

Out[116...

```
Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',  
      'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',  
      'Metascore', 'rating_cat', 'temp'],  
      dtype='object')
```

In [117...

```
data['Genre'].dtype
```

Out[117...

```
dtype('O')
```

In [118...

```
len (data[data['Genre'].str.contains ('Actions',case=False)])
```

Out[118...

```
0
```

22.find unique values from genre

In [119...

```
data.columns
```

Out[119...

```
Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',  
      'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',  
      'Metascore', 'rating_cat', 'temp'],  
      dtype='object')
```

In [120...

```
data['Genre']
```

Out[120... 0 Action,Adventure,Sci-Fi
1 Adventure,Mystery,Sci-Fi
2 Horror,Thriller
3 Animation,Comedy,Family
4 Action,Adventure,Fantasy
...
993 Action,Adventure,Horror
994 Comedy
996 Horror
997 Drama,Music,Romance
999 Comedy,Family,Fantasy
Name: Genre, Length: 838, dtype: object

```
In [121... list1=[]  
for value in data['Genre']:  
    list1.append (value.split(','))
```

```
In [123... one_d=[]  
for item in list1:  
    for item1 in item:  
        one_d.append(item1)
```

```
In [124... one_d
```

Out[124... ['Action',
'Adventure',
'Sci-Fi',
'Adventure',
'Mystery',
'Sci-Fi',
'Horror',
'Thriller',
'Animation',
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'Adventure',
'Sci-Fi',
'Action',
'Comedy',
'Action',
'Crime',
'Drama',
'Biography',
'Drama',
'Comedy',
'Drama',
'Fantasy',
'Romance',
'Action',
'Adventure',
'Sci-Fi',
'Comedy',
'Comedy',
'Drama',
'Animation',
'Action',
'Adventure',
'Animation',
'Adventure',
'Comedy',
'Drama',
'Romance',
'Drama',
'Mystery',
'Thriller',
'Drama',
'History',
'Thriller',
'Animation',

'Adventure',
'Comedy',
'Action',
'Adventure',
'Sci-Fi',
'Adventure',
'Comedy',
'Action',
'Thriller',
'Comedy',
'Music',
'Animation',
'Adventure',
'Comedy',
'Action',
'Adventure',
'Crime',
'Comedy',
'Drama',
'Horror',
'Adventure',
'Family',
'Fantasy',
'Drama',
'Action',
'Drama',
'Thriller',
'Drama',
'Action',
'Horror',
'Romance',
'Action',
'Drama',
'Fantasy',
'Action',
'Crime',
'Drama',
'Drama',
'Fantasy',
'Romance',
...]

In [143...

```
uni_list=[]  
for item in one_d:  
    if item not in uni_list:  
        uni_list.append(item)
```

In [144...

```
uni_list
```

```
Out[144... ['Action',
            'Adventure',
            'Sci-Fi',
            'Mystery',
            'Horror',
            'Thriller',
            'Animation',
            'Comedy',
            'Family',
            'Fantasy',
            'Drama',
            'Music',
            'Biography',
            'Romance',
            'History',
            'Western',
            'Crime',
            'War',
            'Musical',
            'Sport']
```

23 How many Films of Each Genre Were Made?

```
In [142... one_d=[]
for item in list1:
    for item1 in item:
        one_d.append(item1)
```

```
In [130... uni_list
```

```
Out[130... ['Action',
            'Adventure',
            'Sci-Fi',
            'Mystery',
            'Horror',
            'Thriller',
            'Animation',
            'Comedy',
            'Family',
            'Fantasy',
            'Drama',
            'Music',
            'Biography',
            'Romance',
            'History',
            'Western',
            'Crime',
            'War',
            'Musical',
            'Sport']
```

```
In [137... from collections import Counter
Counter(one_d)
```

```
Out[137... Counter({'Drama': 419,
                    'Action': 277,
                    'Comedy': 250,
                    'Adventure': 244,
                    'Thriller': 148,
                    'Crime': 126,
                    'Romance': 120,
                    'Sci-Fi': 107,
                    'Fantasy': 92,
                    'Horror': 87,
                    'Mystery': 86,
                    'Biography': 67,
                    'Family': 48,
                    'Animation': 45,
                    'History': 25,
                    'Music': 15,
                    'Sport': 15,
                    'War': 10,
                    'Musical': 5,
                    'Western': 4})
```

In []: